



School of Future Tech

Case Study Report

on

**Smart Dialect & Accent Phonetic Training
Workshop**

Case Study No.: 150

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1. Introduction

The Smart Dialect & Accent Phonetic Training Workshop is a web-based application developed using React JS. The project is designed to help users improve their pronunciation skills and understand different English accents through interactive speech analysis. The application allows users to practice speaking, compare their pronunciation with reference phrases, and receive instant feedback. By using modern browser technologies, the system provides a simple and effective learning environment without requiring any additional software installation.

2. Problem Statement

Many students and language learners face difficulties in understanding and speaking different English accents. Traditional pronunciation training often requires personal tutors or expensive language-learning software. There is a need for an accessible platform that allows users to practice pronunciation independently. This project addresses this issue by providing a browser-based solution that helps users analyze and improve their speaking skills through speech recognition and pronunciation comparison.

3. Objectives

The objectives of the project are:

- To develop a pronunciation training application using React JS.
- To support multiple English dialects and accents.
- To provide real-time speech recognition.
- To analyze user pronunciation accuracy.
- To visualize speech through audio waveforms.
- To store user progress locally.
- To create an interactive and user-friendly learning platform.

4. Technology Stack

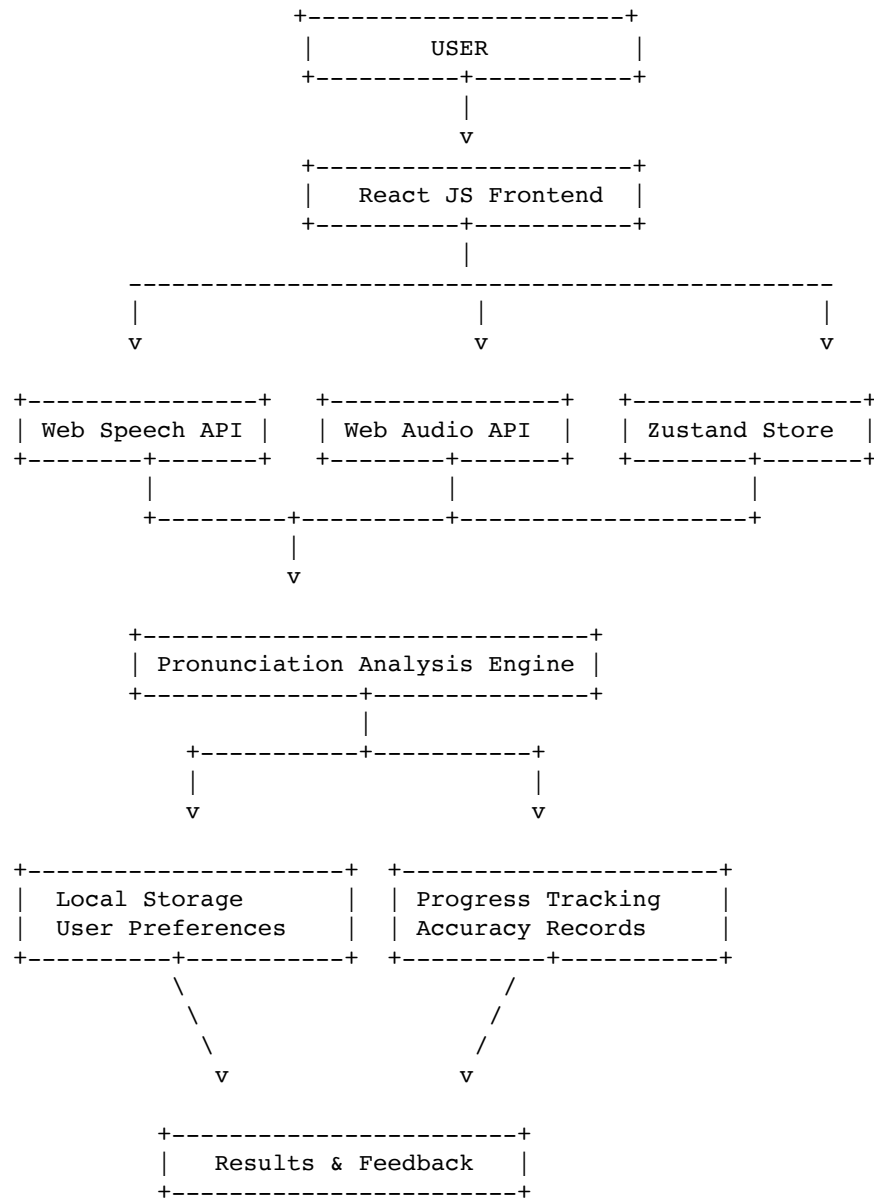
The project uses the following technologies:

- React JS for frontend development.
- Vite as the build tool and development server.
- React Router DOM for page navigation.
- Zustand for state management.
- Web Audio API for audio processing.
- Web Speech API for speech recognition.
- Local Storage for saving user data.
- CSS for styling and responsive design.

5. System Design

The Smart Dialect & Accent Phonetic Training Workshop follows a frontend-based architecture developed using React JS. The application runs completely inside the browser and does not require a backend server or database. All speech processing, pronunciation analysis, and progress tracking are handled on the client side.

The architecture of the system is shown below:



The user interacts with the React JS frontend where speech input is collected through the microphone. The Web Speech API converts speech into text, while the Web Audio API processes the audio signal and generates waveform data. Zustand manages application state and coordinates communication between modules. User progress and settings are stored using Local Storage. Finally, the system generates feedback and displays pronunciation results.

5.1 User Interface Layer

The user interface is developed using React JS components. It provides all controls required for speech practice, dialect selection, and result visualization. The interface is responsive and can be accessed from desktops and laptops without additional software installation.

5.2 Speech Recognition Layer

The Speech Recognition Layer is responsible for converting spoken words into text. When a user speaks through the microphone, the system captures the audio and sends it to the browser's speech recognition service. The recognized text is then compared with the target phrase.

5.3 Audio Processing Layer

The Audio Processing Layer captures microphone input and generates waveform information. This layer helps users visualize their speech patterns and provides a better understanding of pronunciation differences.

5.4 State Management Layer

Zustand is used for global state management. It stores information such as selected dialect, microphone status, current phrase, recognition results, and user progress. State management helps maintain consistency throughout the application.

5.5 Data Storage Layer

The application uses Local Storage to save user preferences and performance records. Since no backend database is used, Local Storage acts as a lightweight persistence mechanism. User information remains available even after refreshing the browser.

5.6 Result Generation Layer

The final layer generates pronunciation feedback based on speech recognition results and comparison logic. Users receive information about pronunciation accuracy, progress, and areas that need improvement.

6. Features

The main features of the application include:

- Real-time speech recognition.
- Multiple accent and dialect support.
- Pronunciation accuracy analysis.
- Interactive waveform visualization.
- Progress tracking system.
- User-friendly interface.
- Browser-based operation.
- Local storage support.
- Responsive design for different devices.
- Fast and efficient performance.

7. Implementation Details

The project follows a modular development approach.

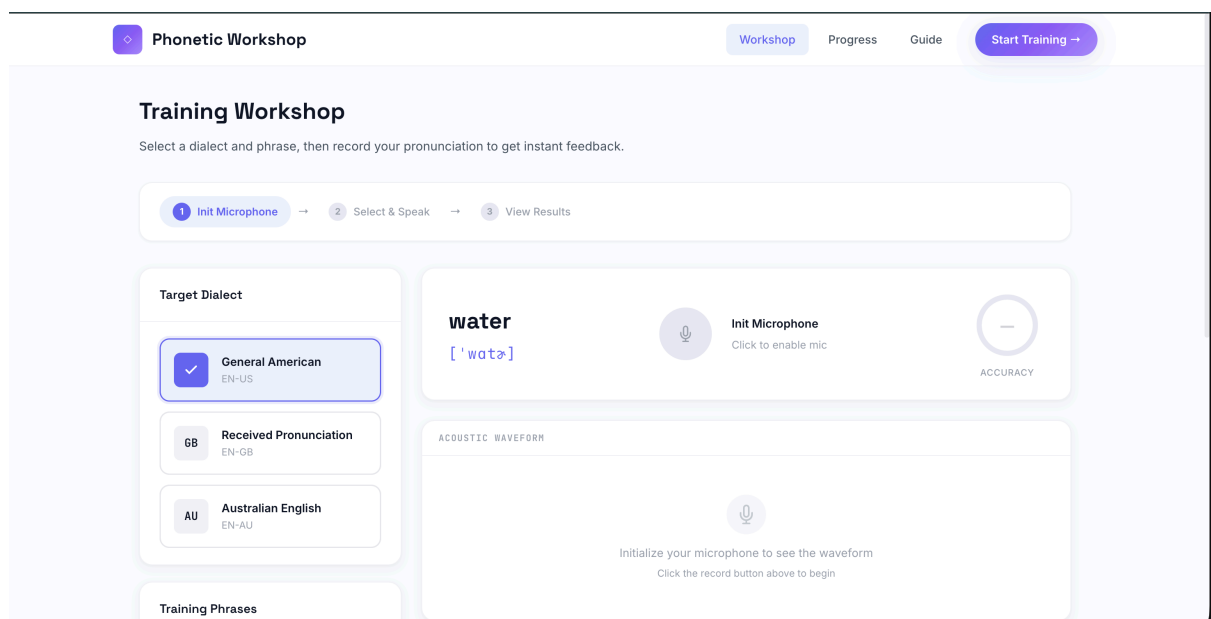
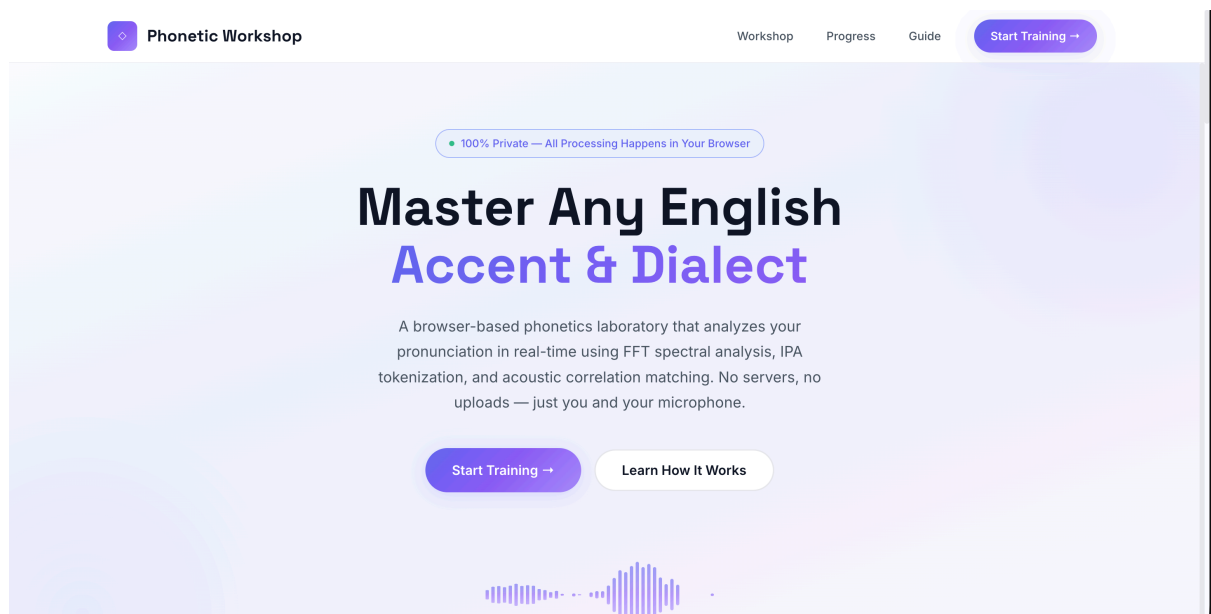
Implementation steps:

1. Create React project using Vite.
2. Configure routing.
3. Build reusable components.
4. Integrate speech recognition.
5. Implement audio processing.
6. Add state management.
7. Create progress tracking system.
8. Design user interface.
9. Test all functionalities.

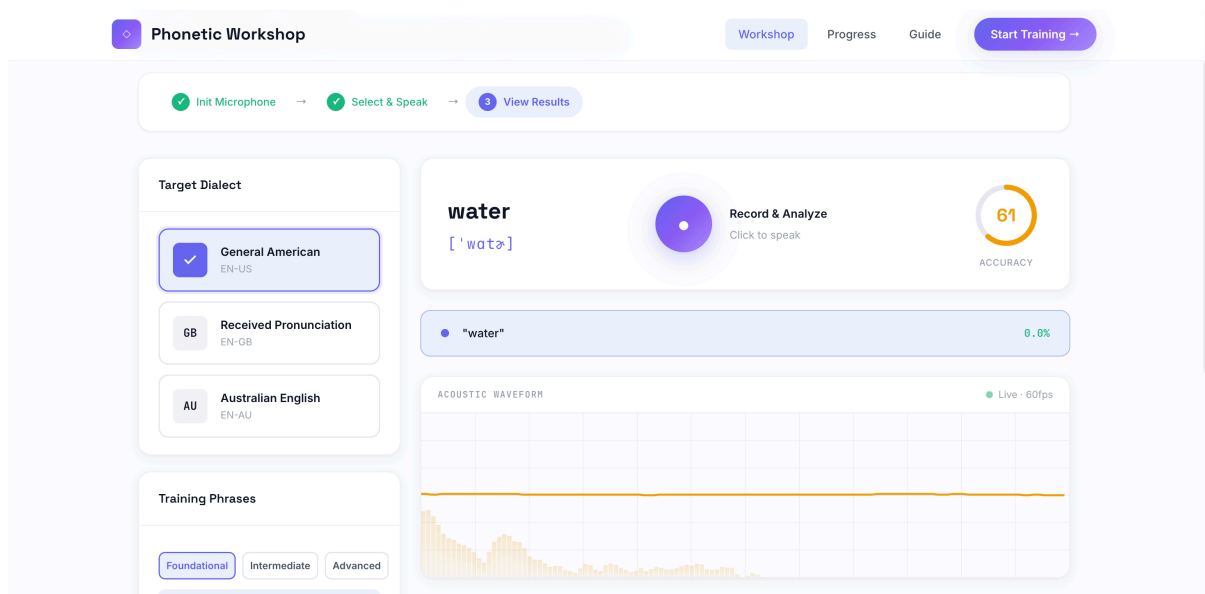
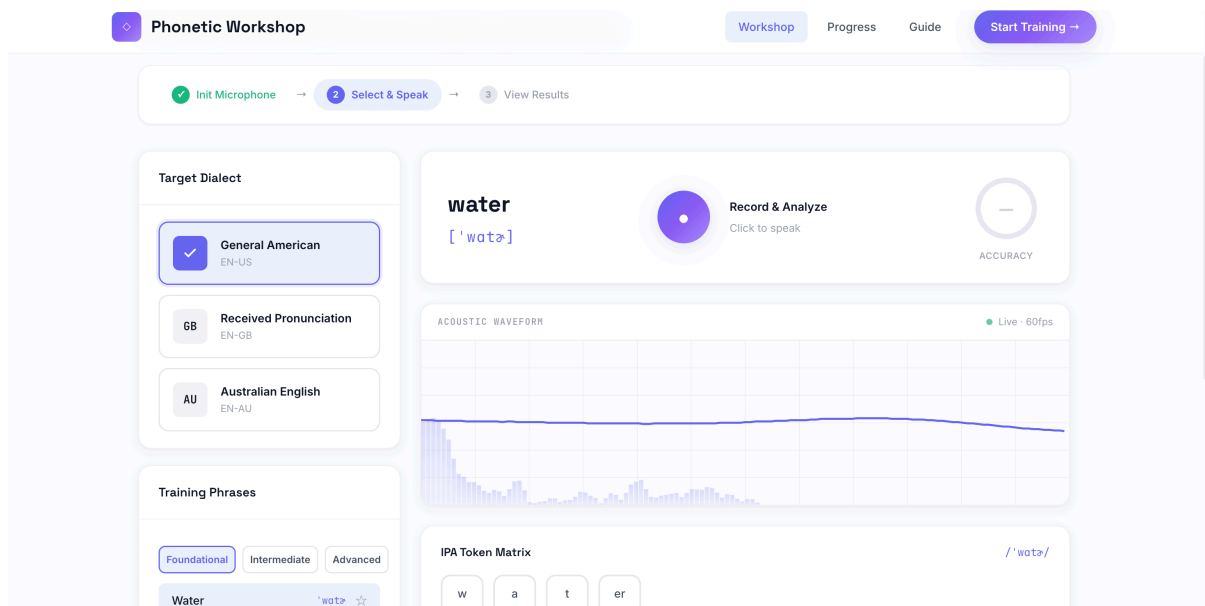
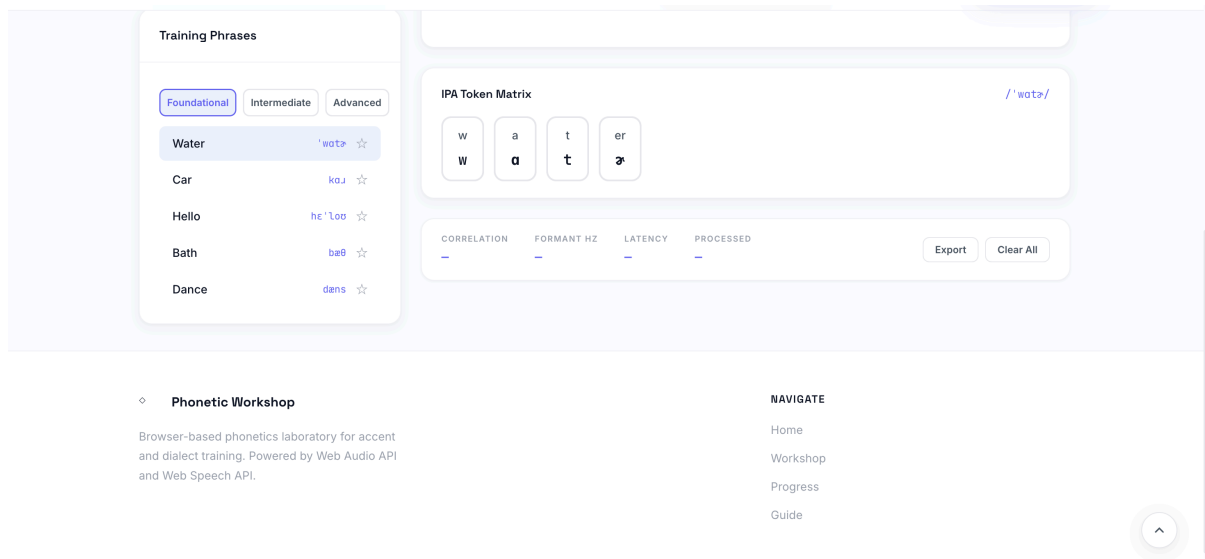
8. Screenshots

This section contains screenshots of the application interface including:

- Home Page
- Workshop Page
- Speech Recognition Interface
- Pronunciation Analysis Screen
- Progress Tracking Dashboard



- Guide Page



FoundationalIntermediateAdvanced

Water'wɒtə'☆

Carkɑː☆

Helloheɪˈləʊ☆

Bathbæθ☆

Dancedɑːns☆

IPA Token Matrix

wW71%

aɑ73%

tɪ49%

erɜː65%

CORRELATION0.6129

FORMANT HZ375

LATENCY5.33ms

PROCESSED230.0 KB

ExportClear All

Phonetic WorkshopWorkshopProgressGuideStart Training →

Your Progress

Track your accent training journey and improvement over time.

#3Total Sessions

%67%Average Accuracy

★74%Best Score

🌐AmericanMost Practiced

Recent Accuracy Scores

water

water

water

Session Historu

Phonetic WorkshopWorkshopProgressGuideStart Training →

Phonetic Training Guide

Learn the fundamentals of phonetics, understand IPA notation, and discover the key differences between English dialects. This guide will help you get the most out of your training sessions.

What is the International Phonetic Alphabet (IPA)?

The International Phonetic Alphabet (IPA) is a standardized system of phonetic notation that represents the sounds of spoken language. Unlike regular spelling, which can be ambiguous (think "through", "though", "tough"), IPA gives every distinct sound its own unique symbol.

For example, the English word "water" is spelled the same everywhere, but pronounced very differently across dialects:

USAmerican English: ['wɒtə] — with a rhotic 'r' sound (ə) and an open 'a' vowel (ɑ)

GBBritish English (RP): ['wɔːtə] — non-rhotic (no 'r' at end), with a rounded 'o' vowel (ɑː)

AUAustralian English: ['wɒːtə] — similar to British but with a more fronted 'o' vowel

9. Results

The application was tested under different usage scenarios.

Testing covered:

- Speech recognition functionality.
- Navigation between pages.
- Local storage operations.
- Audio visualization.
- User interface responsiveness.

Results showed that the application successfully captures speech, analyzes pronunciation, and stores user progress.

10. Conclusion

The Smart Dialect & Accent Phonetic Training Workshop is a useful educational application developed using React JS and modern browser technologies. The project successfully demonstrates speech recognition, pronunciation analysis, audio visualization, and progress tracking within a browser environment.

The application provides users with an effective platform for accent training and pronunciation improvement while showcasing practical implementation of React JS concepts learned during the semester.

11. References

1. React JS Official Documentation
2. Vite Official Documentation
3. React Router DOM Documentation
4. Zustand Documentation
5. Web Audio API Documentation
6. Web Speech API Documentation
7. MDN Web Docs
8. ITM Skills University React JS Course Material